

Core dumped when invoking TFLite model converted using latest nightly TFLite converter (2.4.0dev2020929) #43661

Closed

yousef-xailient opened on Sep 30, 2020 · edited by yousef-xailient

Edits ...

System information

- OS Platform and Distribution (e.g., Linux Ubuntu 16.04): **Linux Ubuntu 20.04**
- TensorFlow installed from (source or binary): **binary**
- TensorFlow version (or github SHA if from source): **2.4.0.dev20200929**

Command used to run the converter or code if you're using the Python API

```
import tensorflow as tf

import numpy as np

def wrap_frozen_graph(graph_def, inputs, outputs):
    def _imports_graph_def():
        tf.compat.v1.import_graph_def(graph_def, name='')
    wrapped_import = tf.compat.v1.wrap_function(_imports_graph_def, [])
    import_graph = wrapped_import.graph
    return wrapped_import.prune(
        tf.nest.map_structure(import_graph.as_graph_element, inputs),
        tf.nest.map_structure(import_graph.as_graph_element, outputs))

graph_def = tf.compat.v1.GraphDef()
_ = graph_def.ParseFromString(open('minimal_093011.pb', 'rb').read())
dnn_function = wrap_frozen_graph(graph_def, inputs='import/first_graph_input:0',
                                outputs='import_1/second_graph_output/Mean:0')
converter = tf.lite.TFLiteConverter.from_concrete_functions([dnn_function])

converter.experimental_enable_mlir_converter = True
converter.optimizations = [tf.lite.Optimize.DEFAULT]
converter.target_spec.supported_ops = [tf.lite.OpsSet.TFLITE_BUILTINS,
tf.lite.OpsSet.SELECT_TF_OPS]

def representative_dataset_gen():
    image = np.random.randint(low=0, high=255, size=(1, 480, 640, 3), dtype='uint8')
    yield [image]

converter.representative_dataset = representative_dataset_gen
converter.inference_input_type = tf.uint8
```

```
converter.inference_output_type = tf.uint8
```

```
model = converter.convert()
```

Link to Google Colab Notebook

https://colab.research.google.com/drive/1U8UVD16lIs1zKjfpFc7hrr3jAo-0eh_i?usp=sharing



Also, please include a link to the saved model or GraphDef

<https://drive.google.com/file/d/1Hvr9hfvaxj3sBi0D0U0iAAe1kEaiJJWB/view?usp=sharing>



Failure details

The conversion is successful in that it generates a tflite graph. However, when I invoke the graph, I get a core dump error:

[1] 511859 abort (core dumped) python src/reproduce_minimal_tflite_test.py

Code used to invoke the graph. Also included in Colab notebook linked above.

```
image = np.random.randint(low=0, high=255, size=(1, 480, 640, 3), dtype='uint8')
```

```
tflite_model = tf.lite.Interpreter('models/minimal_093011.tflite')  
tflite_model.allocate_tensors()
```

```
input_details = tflite_model.get_input_details()  
tflite_model.set_tensor(input_details[0]['index'], image)  
tflite_model.invoke()
```



Traceback

```
#0  __GI_raise (sig=sig@entry=6) at ../sysdeps/unix/sysv/linux/raise.c:50  
#1  0x00007ffff7dc0859 in __GI_abort () at abort.c:79  
#2  0x00007ffffb9386e42 in tflite::QuantizeMultiplierSmallerThanOneExp(double, int*, int*)  
()  
    from /home/yousef/miniconda3/envs/tf2.3/lib/python3.7/site-  
packages/tensorflow/lite/python/interpreter_wrapper/_pywrap_tensorflow_interpreter_wrapper.s  
#3  0x00007ffffb9158090 in void tflite::ops::builtin::comparisons::(anonymous  
namespace)::ComparisonQuantized<signed char, &(bool tflite::reference_ops::GreaterFn<int>  
(int, int))>(TfLiteTensor const*, TfLiteTensor const*, TfLiteTensor*, bo  
ol) () from /home/yousef/miniconda3/envs/tf2.3/lib/python3.7/site-  
packages/tensorflow/lite/python/interpreter_wrapper/_pywrap_tensorflow_interpreter_wrapper.s  
#4  0x00007ffffb9158b7e in tflite::ops::builtin::comparisons::(anonymous  
namespace)::GreaterEval(TfLiteContext*, TfLiteNode*) ()  
    from /home/yousef/miniconda3/envs/tf2.3/lib/python3.7/site-  
packages/tensorflow/lite/python/interpreter_wrapper/_pywrap_tensorflow_interpreter_wrapper.s  
#5  0x00007ffffb9369713 in tflite::Subgraph::Invoke() () from  
/home/yousef/miniconda3/envs/tf2.3/lib/python3.7/site-  
packages/tensorflow/lite/python/interpreter_wrapper/_pywrap_tensorflow_interpreter_wrapper.s  
#6  0x00007ffffb936c1f0 in tflite::Interpreter::Invoke() () from  
/home/yousef/miniconda3/envs/tf2.3/lib/python3.7/site-  
packages/tensorflow/lite/python/interpreter_wrapper/_pywrap_tensorflow_interpreter_wrapper.s  
#7  0x00007ffffb90f7548 in tflite::interpreter_wrapper::InterpreterWrapper::Invoke() ()  
    from /home/yousef/miniconda3/envs/tf2.3/lib/python3.7/site-  
packages/tensorflow/lite/python/interpreter_wrapper/_pywrap_tensorflow_interpreter_wrapper.s  
#8  0x00007ffffb90eb6ee in  
pybind11::cpp_function::initialize<pybind11_init__pywrap_tensorflow_interpreter_wrapper(pybi
```



```

lambda(tflite::interpreter_wrapper::InterpreterWrapper&)#6}, pybind11::object,
tflite::interpreter_wrap
per::InterpreterWrapper&, pybind11::name, pybind11::is_method, pybind11::sibling>
(pybind11_init__pywrap_tensorflow_interpreter_wrapper(pybind11::module&))::
{lambda(tflite::interpreter_wrapper::InterpreterWrapper&)#6}&&, pybind11::object (*
)(tflite::interpreter_wrapper::InterpreterWrapper&), pybind11::name const&,
pybind11::is_method const&, pybind11::sibling const&)::
{lambda(pybind11::detail::function_call&)#3}::_FUN(pybind11::detail::function_call) ()
    from /home/yousef/miniconda3/envs/tf2.3/lib/python3.7/site-
packages/tensorflow/lite/python/interpreter_wrapper/_pywrap_tensorflow_interpreter_wrapper.s
#9 0x00007ffffb90ecb39 in pybind11::cpp_function::dispatcher(_object*, _object*,
_object*) ()
    from /home/yousef/miniconda3/envs/tf2.3/lib/python3.7/site-
packages/tensorflow/lite/python/interpreter_wrapper/_pywrap_tensorflow_interpreter_wrapper.s
#10 0x000055555556b9914 in _PyMethodDef_RawFastCallKeywords (method=0x555555694b100,
self=0x7ffffbb8c9270, args=0x7ffffaf04dd98, nargs=<optimised out>, kwnames=<optimised out>)
    at /tmp/build/80754af9/python_1598874792229/work/Objects/call.c:693
#11 0x000055555556b9a31 in _PyCFunction_FastCallKeywords (func=0x7ffffc08de460, args=
<optimised out>, nargs=<optimised out>, kwnames=<optimised out>) at
/tmp/build/80754af9/python_1598874792229/work/Objects/call.c:732
#12 0x0000555555572639e in call_function (kwnames=0x0, oparg=<optimised out>, pp_stack=
<synthetic pointer>) at /tmp/build/80754af9/python_1598874792229/work/Python/ceval.c:4619
#13 _PyEval_EvalFrameDefault (f=<optimised out>, throwflag=<optimised out>) at
/tmp/build/80754af9/python_1598874792229/work/Python/ceval.c:3093
#14 0x000055555556b8e7b in function_code_fastcall (globals=<optimised out>, nargs=1, args=
<optimised out>, co=<optimised out>) at
/tmp/build/80754af9/python_1598874792229/work/Objects/call.c:283
#15 _PyFunction_FastCallKeywords (func=<optimised out>, stack=0x7ffff6d615c0, nargs=1,
kwnames=<optimised out>) at
/tmp/build/80754af9/python_1598874792229/work/Objects/call.c:408
#16 0x00005555555721740 in call_function (kwnames=0x0, oparg=<optimised out>, pp_stack=
<synthetic pointer>) at /tmp/build/80754af9/python_1598874792229/work/Python/ceval.c:4616
#17 _PyEval_EvalFrameDefault (f=<optimised out>, throwflag=<optimised out>) at
/tmp/build/80754af9/python_1598874792229/work/Python/ceval.c:3110
#18 0x00005555555668829 in _PyEval_EvalCodeWithName (_co=0x7ffff6cfa1e0, globals=
<optimised out>, locals=<optimised out>, args=<optimised out>, argcount=<optimised out>,
kwnames=0x0, kwargs=0x0, kwcount=0, kwstep=2, defs=0x0, defcount=0,
    kwdefs=0x0, closure=0x0, name=0x0, qualname=0x0) at
/tmp/build/80754af9/python_1598874792229/work/Python/ceval.c:3930
#19 0x00005555555669714 in PyEval_EvalCodeEx (_co=<optimised out>, globals=<optimised
out>, locals=<optimised out>, args=<optimised out>, argcount=<optimised out>, kws=
<optimised out>, kwcount=0, defs=0x0, defcount=0, kwdefs=0x0,
    closure=0x0) at /tmp/build/80754af9/python_1598874792229/work/Python/ceval.c:3959
#20 0x0000555555566973c in PyEval_EvalCode (co=<optimised out>, globals=<optimised out>,
locals=<optimised out>) at
/tmp/build/80754af9/python_1598874792229/work/Python/ceval.c:524
#21 0x00005555555780f14 in run_mod (mod=<optimised out>, filename=<optimised out>,
globals=0x7ffff6dcac30, locals=0x7ffff6dcac30, flags=<optimised out>, arena=<optimised
out>)
    at /tmp/build/80754af9/python_1598874792229/work/Python/pythonrun.c:1035
#22 0x0000555555578b331 in PyRun_FileExFlags (fp=0x55555558c3100, filename_str=<optimised
out>, start=<optimised out>, globals=0x7ffff6dcac30, locals=0x7ffff6dcac30, closeit=1,
flags=0x7ffffffffffdd80)
    at /tmp/build/80754af9/python_1598874792229/work/Python/pythonrun.c:988
#23 0x0000555555578b523 in PyRun_SimpleFileExFlags (fp=0x55555558c3100, filename=<optimised
out>, closeit=1, flags=0x7ffffffffffdd80) at
/tmp/build/80754af9/python_1598874792229/work/Python/pythonrun.c:429
#24 0x0000555555578c655 in pymain_run_file (p_cf=0x7ffffffffffdd80, filename=0x55555558c2870
L"src/reproduce_minimal_tflite_test.py", fp=0x55555558c3100) at
/tmp/build/80754af9/python_1598874792229/work/Modules/main.c:462
#25 pymain_run_filename (cf=0x7ffffffffffdd80, pymain=0x7ffffffffffde90) at
/tmp/build/80754af9/python_1598874792229/work/Modules/main.c:1652
#26 pymain_run_python (pymain=0x7ffffffffffde90) at
/tmp/build/80754af9/python_1598874792229/work/Modules/main.c:2913
#27 pymain_main (pymain=0x7ffffffffffde90) at

```

```
/tmp/build/80754af9/python_1598874792229/work/Modules/main.c:3460
#28 0x00005555578c77c in _Py_UnixMain (argc=<optimised out>, argv=<optimised out>) at
/tmp/build/80754af9/python_1598874792229/work/Modules/main.c:3495
#29 0x00007ffff7dc20b3 in __libc_start_main (main=0x555555649c90 <main>, argc=2,
argv=0x7ffffffffffdf8, init=<optimised out>, fini=<optimised out>, rtld_fini=<optimised
out>, stack_end=0x7ffffffffffdfe8) at ../csu/libc-start.c:308
#30 0x000055555730ff0 in _start () at ../sysdeps/x86_64/elf/start.S:103
```

 **yousef-xailient** added **TFLiteConverter** on Sep 30, 2020

 **google-ml-butler** assigned [amahendrakar](#) on Sep 30, 2020

 **yousef-xailient** mentioned this on Sep 30, 2020

✓ [Error converting from TensorFlow frozen graph to TFLite using TF1.15 #43660](#)

 amahendrakar on Sep 30, 2020

Contributor ...

@yousef-xailient,

I do not have access to the Colab notebook and the saved model file. Could you please grant the required permissions to view the files. Thanks!

 **amahendrakar** added **comp:lite** **stat:awaiting resp...** **TF 2.3** **type:support** on Sep 30, 2020

 yousef-xailient on Oct 1, 2020

Author ...

@amahendrakar

My bad. You can try now.

 **tensorflowbutler** removed **stat:awaiting resp...** on Oct 2, 2020

 amahendrakar on Oct 5, 2020

Contributor ...

Was able to reproduce the issue. Session crashes on running the code with TF v2.3 and TF-nightly. Please find the gist of it [here](#). Thanks!

 **amahendrakar** added **type:bug** and removed **type:support** on Oct 5, 2020

 **amahendrakar** unassigned [amahendrakar](#) on Oct 5, 2020

8 remaining items

Load more

karimnosseir on Jun 26, 2021

Contributor ...

Hi, is this still on going issue ?

Thanks

dx-xp-team on Jul 29, 2021

...

Hi,

I have the exact same problem and coredump.

I have searched everywhere and I did not found any solution to this issue.

The problem is still reproducible on latest version (2.7.0-dev20210729) through the Collab link :

<https://colab.research.google.com/gist/amahendrakar/9253e439795b5ccf487ada3ab385bb90/43661-tf-nightly.ipynb>

I would be nice to have some feedback/investigation why the crash occurs...

Thanks for your support

  **mohantym** added **2.6.0** on Oct 24, 2021

  **mohantym** self-assigned this on Mar 3, 2022

mohantym on Mar 3, 2022

Contributor ...

I was able to replicate in [2.8 version](#) too.

  **karimnosseir** assigned [abattery](#) and unassigned [karimnosseir](#) on Mar 4, 2022

karimnosseir on Mar 4, 2022

Contributor ...

That looks invalid values from quantization params.

Jaesung can you please have a look

Thanks

abattery on Mar 4, 2022

Contributor ...

Thanks. [@sngyhan](#) fyi

mihaimaruseac on Mar 5, 2022

Contributor ...

Can you please tag me too in the fix?



mohantym removed their assignment on Mar 22, 2022

mihaimaruseac on Mar 23, 2022

Contributor ...

This should be fixed by [a989426](#)



mihaimaruseac closed this as completed on Mar 23, 2022

google-ml-butler bot on Mar 23, 2022

...

Are you satisfied with the resolution of your issue?

[Yes](#)

[No](#)



judeharis mentioned this on Mar 16, 2023

👍 Problems with converted 8-bit TFLite models of CycleGAN and running inference (specially allocating tensors) #59922

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Assignees



abattery

Labels

2.6.0

TF 2.3

TFLiteConverter

comp:lite

stat:awaiting tensorflow

type:bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

 Code with agent mode

▼

No branches or pull requests

Participants



+4